Engine Model(s): PT6A

73-10-04

FLOW DIVIDER AND STARTING CONTROL - DESCRIPTION AND OPERATION

1. Description and Operation

A. General (PT6A-21 Engines)

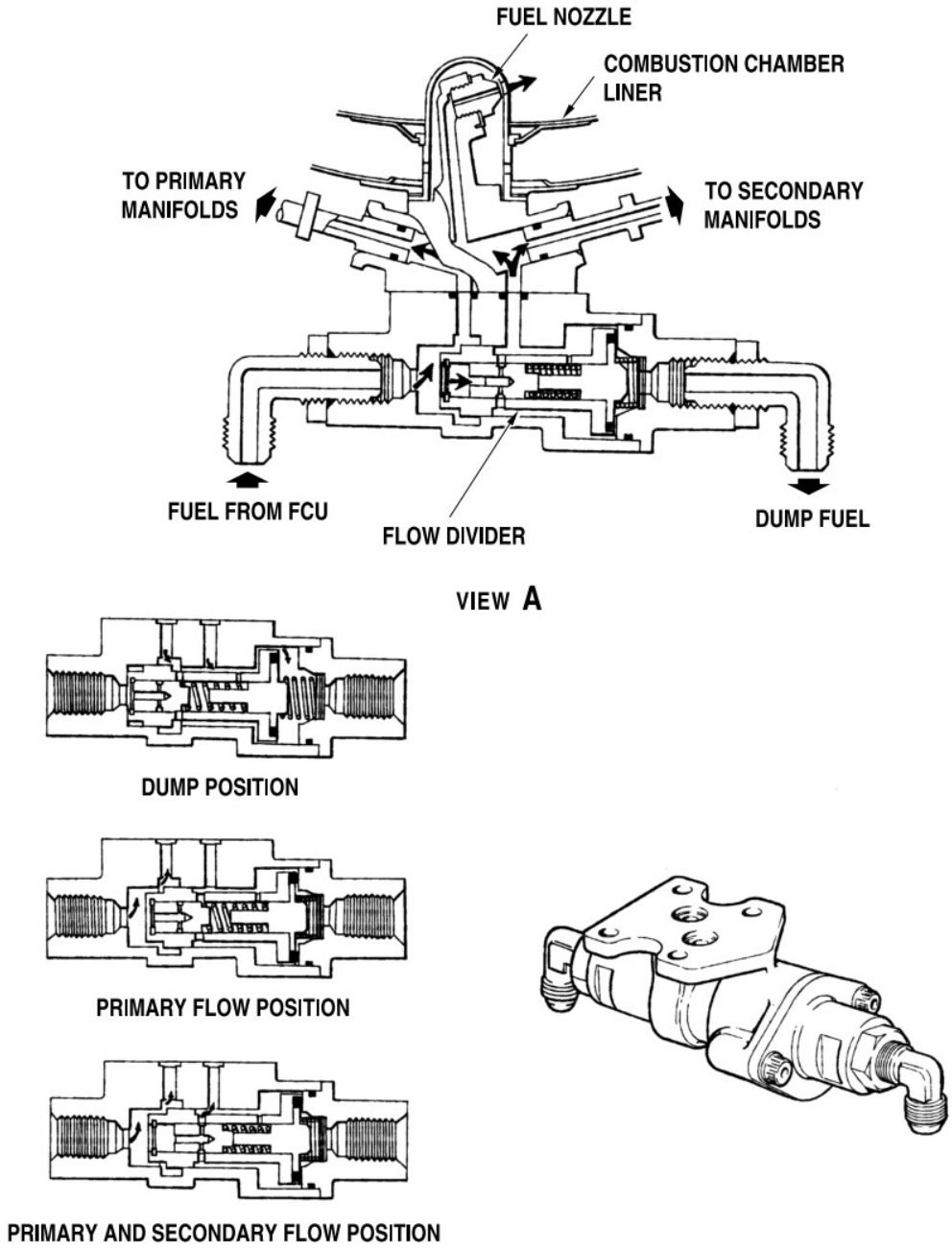
There are two alternative models of flow divider; one incorporating a dump valve and the other a purge valve.

The flow divider and dump valve or purge vlave assembly is mountedon the fuel inlet manifold adapter located at the 6 o'clock position on the gas generator case. The flow divider schedules the metered fuel from the FCU between the primary and secondary fuel manifolds as a function of primary manifold pressure. During engine start-up, metered fuel is delivered initially by primary nozzles, with the secondary nozzles cutting in above a preset value. All nozzles are operative at idle and above.

(1) Flow Divider and (Pre-SB1406) Dump Valve (Ref. Fig. 1)

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Figure 1 Flow Divider and Dump Valve



Pre-SB1406, PT6A-21

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The flow divider and dump valve divides the fuel flow from the fuel control unit between the primary and secondary fuel manifolds during engine start-up and operation, and dumps residual fuel from the manifolds at engine shutdown.

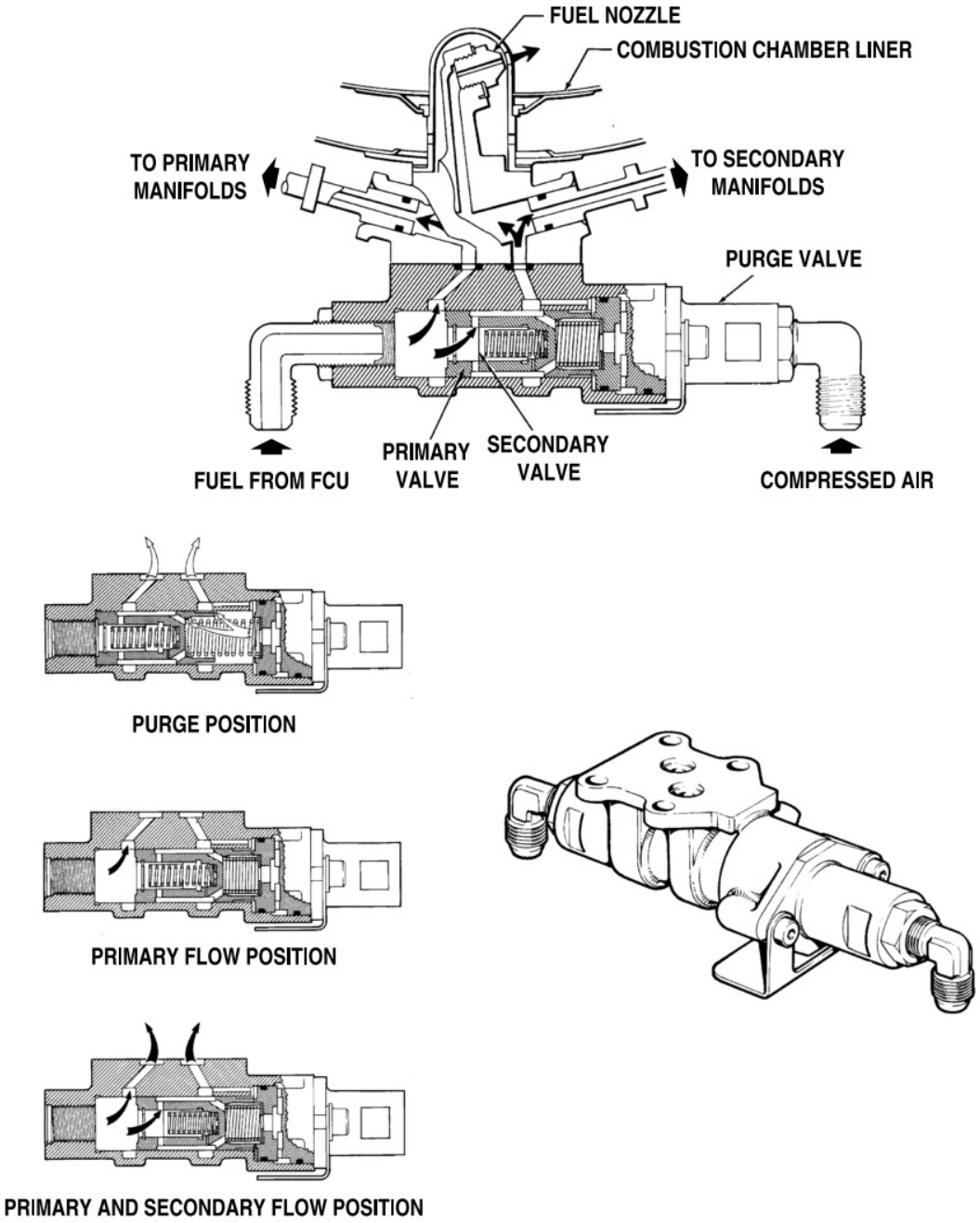
The unit consists of a spring-loaded transfer valve operating within a spring-loaded, ported and slotted dump valve plunger. The transfer valve and plunger are housed in a single-unit body. Ports are provided for fuel inlet, primary and secondary fuel manifold flows, and residual fuel dump. The unit is fitted with two elbow connectors for the fuel inlet and dump lines.

When the fuel cutoff valve in the FCU closes during engine shutdown, an operating spring in the flow divider and dump valve overcomes the fuel inlet pressure and moves a piston within the unit body to block the fuel inlet port. This movement of the piston connects both primary and secondary manifolds to the dump valve port and allows residual fuel in the manifold to drain overboard, or to an airframe drain collection point, as applicable.

(2) Flow Divider and (Post-SB1406) Purge Valve (Ref. Fig. 2)

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Figure 2 Flow Divider and Purge Valve



Post-SB1406, PT6A-21

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The flow divider and purge valve divides the fuel flow from the fuel control unit between the primary and secondary fuel manifoldsduring engine start-up and operation, and flushes residual fuel from the manifolds into the combustion chamber at engine shutdown.

The unit consists of a primary valve, an independent secondary valve and an air purge valve. The primary valve is housed in the main body of the unit, and controls primary manifold fuel flow. The secondary valve is isolated within the primary valve, and controls secondary manifold fuel flow. Both the primary and secondary valves are spring-loaded and are shimmed to give precise cracking pressures.

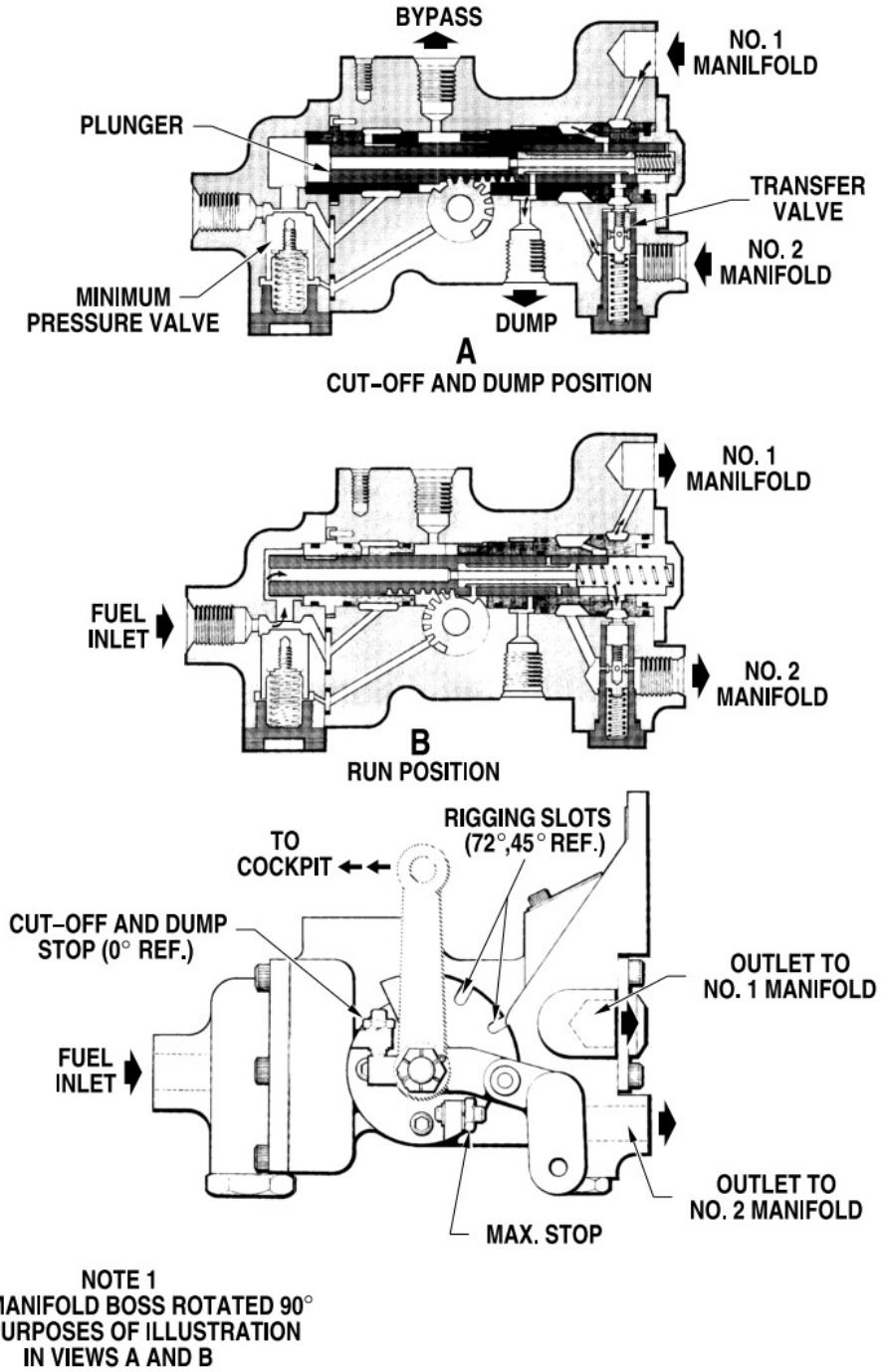
A soft-seat check valve is located inside the purge valve to prevent fuel from entering the air purge accumulator, which is charged with compressed air. On engine shutdown, the valve opens and allows compressed air to flush the residual fuel from the manifolds into the combustion chamber where it is ignited and burned off.

Ports are provided for fuel inlet, primary and secondary fuel manifold flows and residual fuel purge. Two elbow connectors are fitted to the unit for the fuel inlet and purge lines.

B. Starting Flow Control (PT6A-27/-28 Engines) (Ref. Fig. 3)

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Figure 3 Starting Flow Control - Schematic



PT6A-27 and PT6A-28

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The starting flow control is mounted at the 5 o'clock position on flange G of the accessory gearbox and connected by rigid stainless-steel fuel tubes to the FCU and fuel manifold.

The control consists of a ported plunger sliding in a ported housing. Rotational movement of the input lever is converted to a linear plunger movement through a rack and pinion arrangement. A minimum pressurizing valve, in the control inlet, maintains a minimum pressure in the FCU to ensure correct fuel metering. Two outlet ports are provided, one for the primary manifold and the other for the secondary manifold; both ports are interconnected via a transfer valve. This valve allows the primary manifold to fill initially for engine lightup and, as pressure increases in the control, the valve opens allowing fuel into the secondary manifold.

When the input lever is in the cutoff and dump position (zero degree), the fuel supply to both fuel manifolds is blocked off and directed back to the FCU via bypass tubing; at the same time drain ports are aligned, via porting in the plunger, with the control dump port to allow residual fuel in both primary and secondary manifolds and lines to drain over board. This prevents fuel being coked in the system due to heat absorp tion. Fuel entering the starting control during engine rundown is diverted via the bypass port to the fuel pump inlet.

When the input lever is placed in the RUN position, the outlet port to the primary (No. 1) manifold is uncovered and the bypass port is covered. As the engine accelerates, both fuel flow and manifold pressure increase until the transfer valve opens and the secondary (No. 2) manifold fills. When the secondary manifold is filled, the total flow to the engine is increased by the amount now being delivered through the secondary manifold and the engine accelerates to idle. When the input lever is moved beyond the RUN position (45 degrees or 72 degrees, as applicable) towards the maximum stop (90 degrees) the starting flow control then has no further effect on fuel being metered to the engine. This range is overtravel and is utilized to actuate the engine high-idle requirement via a telescopic interconnecting linkage to the FCU on most airframe installations.

Rigging slots are provided on the input lever cam at the 45-degree and 72-degree RUN positions. One of these rig positions, depending on the airframe installation, is used to rig the system and to identify the cockpit lever position.

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